Recurrence Relations: Solved Examples using the Direct (Iterative) Method

Example 1: $T(n) = T(n-1) + n$

Solution:

$$\begin{aligned}
T(n) &= T(n-1) + n \\
&= T(n-2) + (n-1) + n \\
&= T(n-3) + (n-2) + (n-1) + n \\
&\vdots \\
&= T(1) + 2 + 3 + \cdots + n \\
&= \Theta(1) + \sum_{k=2}^{n} k = \frac{n(n+1)}{2} - 1
\end{aligned}$$

Final Answer: $\boxed{T(n) = \Theta(n^2)}$

Example 2: $T(n) = 2T(n/2) + n$

Assume: $n = 2^k$

$$\begin{aligned}
T(n) &= 2T(n/2) + n \\
&= 2[2T(n/4) + n/2] + n = 4T(n/4) + 2n \\
&= 4[2T(n/8) + n/4] + 2n = 8T(n/8) + 3n \\
&\vdots \\
&= 2^k T(1) + kn = n \cdot T(1) + n \log n
\end{aligned}$$

Final Answer: $\boxed{T(n) = \Theta(n \log n)}$

Example 3: $T(n) = T(n/2) + n$

Assume: $n = 2^k$

$$\begin{aligned}
T(n) &= T(n/2) + n \\
&= T(n/4) + n/2 + n \\
&= T(n/8) + n/4 + n/2 + n \\
&\vdots \\
&= T(1) + n + n/2 + n/4 + \dots \\
&= T(1) + 2n
\end{aligned}$$

Final Answer: $\boxed{T(n) = \Theta(n)}$

Example 4: $T(n) = 3T(n/3) + n$

Assume: $n = 3^k$

$$\begin{aligned} T(n) &= 3T(n/3) + n \\ &= 3[3T(n/9) + n/3] + n = 9T(n/9) + 2n \\ &= 9[3T(n/27) + n/9] + 2n = 27T(n/27) + 3n \\ &\vdots \\ &= 3^k T(1) + kn = n \cdot T(1) + n \log_3 n \end{aligned}$$

Final Answer: $\boxed{T(n) = \Theta(n \log n)}$

Example 5: $T(n) = 2T(n/2) + n \log n$

Assume: $n = 2^k$

$$\begin{aligned} T(n) &= 2T(n/2) + n \log n \\ &= 2[2T(n/4) + (n/2)\log(n/2)] + n \log n \\ &= 4T(n/4) + n \log(n/2) + n \log n \\ &= 8T(n/8) + n \log(n/4) + n \log(n/2) + n \log n \\ &\vdots \\ &= 2^k T(1) + n \sum_{i=0}^{k-1} \log(n/2^i) \end{aligned}$$

Simplify the sum:

$$\sum_{i=0}^{\log n - 1} \log(n/2^i) = \sum_{i=0}^{\log n - 1} (\log n - i) = \frac{\log n(\log n + 1)}{2}$$

Final Answer: $\boxed{T(n) = \Theta(n \log^2 n)}$

Example 6:

$$T(n) = 3T\left(\frac{n}{4}\right) + n$$

and show that $T(n) = \Theta(n)$

Step-by-Step Expansion

We expand the recurrence step by step to find a pattern.

Level 0:

$$T(n) = 3T\left(\frac{n}{4}\right) + n$$

Level 1:

$$= 3\left[3T\left(\frac{n}{4^2}\right) + \frac{n}{4}\right] + n = 3^2 T\left(\frac{n}{4^2}\right) + 3 \cdot \frac{n}{4} + n$$

Level 2:

$$= 3^3 T\left(\frac{n}{4^3}\right) + 3^2 \cdot \frac{n}{4^2} + 3 \cdot \frac{n}{4} + n$$

Continuing this to level k:

$$T(n) = 3^k T\left(\frac{n}{4^k}\right) + n \sum_{i=0}^{k-1} \left(\frac{3}{4}\right)^i$$

Geometric Series

$$\sum_{i=0}^{k-1} \left(\frac{3}{4}\right)^i = \frac{1 - \left(\frac{3}{4}\right)^k}{1 - \frac{3}{4}} = 4\left(1 - \left(\frac{3}{4}\right)^k\right)$$

Substituting back:

$$T(n) = 3^k T\left(\frac{n}{4^k}\right) + 4n\left(1 - \left(\frac{3}{4}\right)^k\right)$$

Base Case We stop when:

$$\frac{n}{4^k} = 1 \Rightarrow 4^k = n \Rightarrow k = \log_4 n$$

Now substitute $k = \log_4 n$:

$$T(n) = 3^{\log_4 n} \cdot T(1) + 4n\left(1 - \left(\frac{3}{4}\right)^{\log_4 n}\right)$$

Simplifying the Expression

Recall:

$$3^{\log_4 n} = n^{\log_4 3} \quad \text{and} \quad \left(\frac{3}{4}\right)^{\log_4 n} = n^{\log_4 (3/4)}$$

So:

$$T(n) = n^{\log_4 3} \cdot T(1) + 4n\left(1 - n^{\log_4 (3/4)}\right)$$

Note:

- $\log_4 3 \approx 0.792 \Rightarrow n^{\log_4 3} = o(n)$
- $n^{\log_4 (3/4)} \to 0$ as $n \to \infty$

Thus:

$$T(n) = o(n) + \Theta(n) = \Theta(n)$$